Theorem 3.6

A polynomial equation $a_n x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \cdots + a_2 x^2 + a_1 x + a_0 = 0$, $(a_n \neq 0)$ is a reciprocal equation if, and only if, one of the following two statements is true:

(i) $a_n = a_0$, $a_{n-1} = a_1$, $a_{n-2} = a_2 \cdots$

(ii) $a_n = -a_0$, $a_{n-1} = -a_1$, $a_{n-2} = -a_2, \cdots$

Proof

Consider the polynomial equation

$$P(x) = a_n x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \cdots + a_2 x^2 + a_1 x + a_0 = 0 . \quad \ldots (1)$$

Replacing x by $\frac{1}{x}$ in (1), we get

$$P\left(\frac{1}{x}\right) = \frac{a_n}{x^n} + \frac{a_{n-1}}{x^{n-1}} + \frac{a_{n-2}}{x^{n-2}} + \cdots + \frac{a_2}{x^2} + \frac{a_1}{x} + a_0 = 0 . \quad \ldots (2)$$

Multiplying both sides of (2) by x^n, we get

$$x^n P\left(\frac{1}{x}\right) = a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + \cdots + a_{n-2}x^2 + a_{n-1}x + a_n = 0 . \quad \ldots (3)$$

Now, (1) is a reciprocal equation $\Leftrightarrow$ $P(x) = \pm\, x^n P\left(\frac{1}{x}\right) \Leftrightarrow$ (1) and (3) are same .

This is possible $\Leftrightarrow$ $\frac{a_n}{a_0} = \frac{a_{n-1}}{a_1} = \frac{a_{n-2}}{a_2} = \cdots = \frac{a_2}{a_{n-2}} = \frac{a_1}{a_{n-1}} = \frac{a_0}{a_n}$.

Let the proportion be equal to λ. Then, we get $\frac{a_n}{a_0} = \lambda$ and $\frac{a_0}{a_n} = \lambda$. Multiplying these equations, we get $\lambda^2 = 1$. So, we get two cases $\lambda = 1$ and $\lambda = -1$.

Case (i) :

$\lambda = 1$ In this case, we have $a_n = a_0$, $a_{n-1} = a_1$, $a_{n-2} = a_2$, $\cdots$.

That is, the coefficients of (1) from the beginning are equal to the coefficients from the end.

Case (ii) :

$\lambda = -1$ In this case, we have $a_n = -a_0$, $a_{n-1} = -a_1$, $a_{n-2} = -a_2$, $\cdots$.

That is, the coefficients of (1) from the beginning are equal in magnitude to the coefficients from the end, but opposite in sign. ■

Note

Reciprocal equations of Type I correspond to those in which the coefficients from the beginning are equal to the coefficients from the end.

For instance, the equation $6x^5 + x^4 - 43x^3 - 43x^2 + x + 6 = 0$ is of type I.

Reciprocal equations of Type II correspond to those in which the coefficients from the beginning are equal in magnitude to the coefficients from the end, but opposite in sign.

For instance, the equation $6x^5 - 41x^4 + 97x^3 - 97x^2 + 41x - 6 = 0$ is of Type II.

Remark

(i) A reciprocal equation cannot have 0 as a solution.

(ii) The coefficients and the solutions are not restricted to be real.

Chapter 3 Theory of Equation.indd 121 31-01-2020 17:34:08